

AQI Prediction System

Sukkur — Karachi — Lahore

Final Project Report

10Pearls Shine Internship Program — Data Science

Prepared by: Bharat Kumar

Mentors: Ms. Umema Ashar, Ms. Hafsa Imtiaz

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1. Introduction

This report covers an AQI (Air Quality Index) prediction project built during the 10Pearls Shine Internship Program, Data Science track. The goal was simple to state and harder to build properly: given live pollution and weather readings for a city, predict what the AQI is right now, and also forecast what it's likely to be over the next three days.

The project covers three cities — Sukkur, Karachi, and Lahore — and is built as an end-to-end pipeline: data collection, feature engineering, model training, automation, and a dashboard, rather than a single notebook that trains one model and stops there.

1.1 What this report does NOT do

In the interest of being honest about the work, this report does not claim results that weren't actually produced by running the project. Sections where real numbers belong are marked clearly and left for the actual training run output — padding a report with invented metrics defeats the point of building something real.

2. Problem Statement and Objectives

- Predict the current AQI for a city from live pollutant and weather readings.
- Forecast AQI for the next 3 days (24h, 48h, and 72h ahead), not just repeat today's number.
- Automate data collection and model retraining so the system stays current on its own.
- Store features and models in Hopsworks, so training and the dashboard always read from the same source.
- Present everything through a dashboard that's actually usable by someone who isn't a data scientist.

3. System Architecture

The project is split into separate scripts, each doing one job. This makes it easier to debug than one big script that does everything — if something breaks, it's usually obvious which stage broke.

Stage	Script	What it does
1	feature_pipeline.py	Fetches live pollution + weather data every hour, computes AQI and features, stores them.
2	backfill.py	Pulls historical pollution data (2 years) to build the initial training set.
3	train_current.py	Trains the model that predicts AQI from live readings, right now.
4	train_forecast.py	Trains three models that forecast AQI 24h / 48h / 72h ahead.
5	GitHub Actions	Runs stage 1 hourly and stages 3–4 daily, automatically.
6	app.py	Streamlit dashboard showing current AQI, forecasts, trends, and comparisons.

3.1 Technology stack

Layer	Technology
Data source	OpenWeather Air Pollution API and Weather API

Feature store	Hopsworks
Model registry	Hopsworks Model Registry
Model training	scikit-learn — Ridge Regression and Random Forest
Automation	GitHub Actions (scheduled workflows)
Dashboard	Streamlit

4. Two Separate Models — and Why

This is the part of the project that took the most rethinking, so it's worth explaining properly rather than glossing over it.

AQI is not really an independent measurement — it's a number calculated FROM pollutant readings like PM2.5 and PM10, using a standard formula. That creates a trap: if you build a model to predict AQI and casually feed it the current AQI value (or something derived from it) as an input, the model isn't learning anything real. It's just seeing a version of the answer and handing it back. That's data leakage, and it makes results look far better than they actually are.

To avoid this, the project uses two separate models with two separate feature sets:

4.1 Current AQI Prediction Model

Predicts AQI from live pollutant and weather readings only — PM2.5, PM10, CO, NO, NO2, O3, SO2, NH3, temperature, humidity, pressure, and wind speed. AQI itself is never used as an input here. Since each row is an independent snapshot in time (not a future prediction), a normal random train/test split is used.

4.2 3-Day Forecast Models

Predicts AQI 24, 48, and 72 hours ahead. This one IS allowed to use past AQI values as features (AQI from 1 hour ago, 3 hours ago, 24 hours ago, plus a rolling average) — this is not leakage, because the model only ever sees values from BEFORE the point it's predicting. This is exactly how normal time-series forecasting works: use the past to estimate the future, never use the target itself.

A time-based train/test split is used here instead of a random one, since a random split could put future rows into training and past rows into testing — which would let the model quietly learn from data it shouldn't have access to yet.

5. Data Collection and Feature Selection

5.1 Cities

Sukkur, Karachi, and Lahore — chosen because they're geographically different (Sukkur is smaller and inland, Karachi is a large coastal city, Lahore is a large inland city), so their AQI patterns aren't identical, which makes the city comparison part of the dashboard meaningful instead of showing three near-identical lines.

5.2 Features used, and why

Feature	Used in	Reason
PM2.5, PM10	Both models	Core pollutants; AQI is directly derived from these.
CO, NO, NO2, O3, SO2, NH3	Both models	Always returned by the API, no missing-data problems.
Temperature, Humidity, Pressure, Wind speed	Current model	Affect how pollution disperses; always available live.

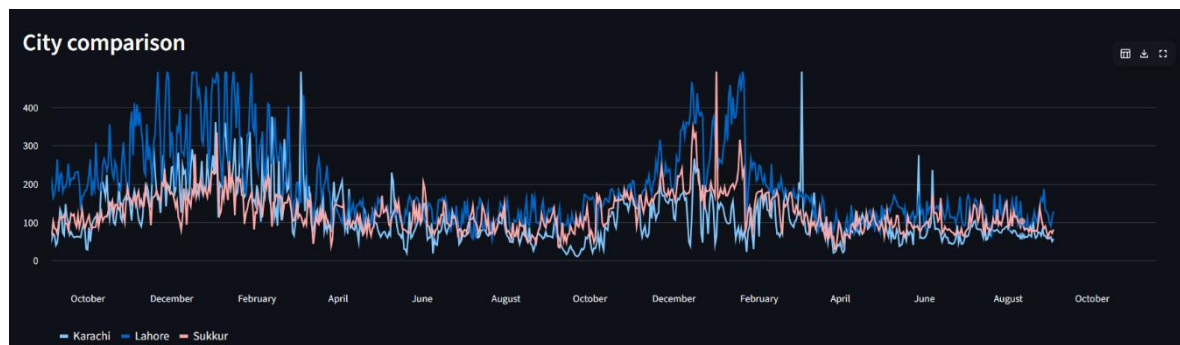
Wind direction, Cloudiness, Rain	Shown on dashboard only	Available live but not in historical backfill — not enough training history to use as model features.
AQI (current value)	Forecast model only	Used as a lag feature to predict the future — not used in the current-prediction model, to avoid leakage.

One limitation worth stating plainly: OpenWeather's free plan only provides historical WEATHER data for about a week back, while historical POLLUTION data goes back much further (to late 2020). This means older, backfilled rows only have pollutant columns filled in — weather columns are empty until the hourly pipeline starts collecting live readings going forward. Because of this, the current-AQI model can only train on the smaller set of rows that actually have complete weather data.

6. Exploratory Data Analysis

Basic EDA was done on the collected dataset before training, to sanity-check the data and look for obvious patterns.

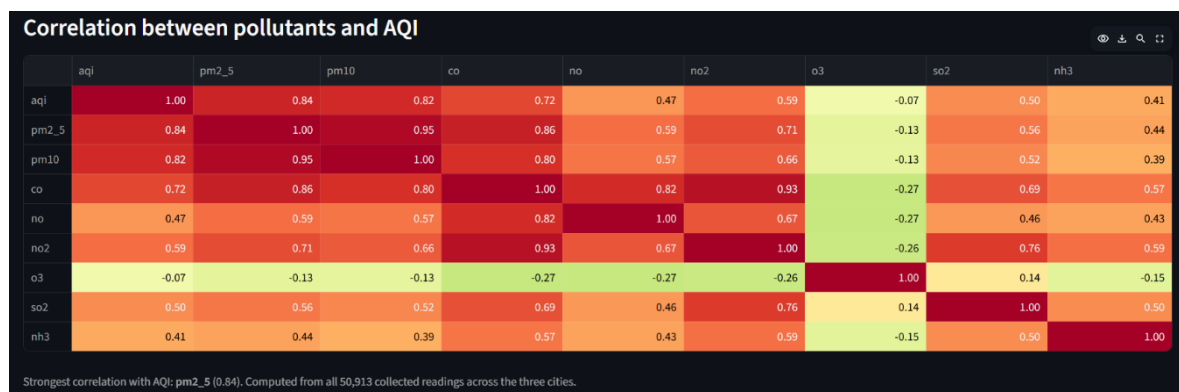
AQI distribution per city



AQI trend over time for each city



Correlation between pollutants and AQI



Notes on what these charts showed:

The AQI trends show noticeable variation across Karachi, Lahore, and Sukkur over time. Some periods show significant increases in AQI, indicating poorer air quality. The city comparison also shows that pollution levels vary between cities and across different seasons. The correlation analysis shows that particulate pollutants, particularly PM2.5 and PM10, have a strong positive relationship with AQI. This means that when particulate matter increases, AQI generally tends to increase as well. Other pollutants such as CO, NO₂, SO₂, and O₃ also contribute to changes in AQI, although their relationships may be weaker or different depending on the dataset. Overall, the EDA suggests that PM2.5 and PM10 are important factors associated with AQI, while AQI also changes over time and differs between cities.

7. Model Training and Results

7.1 Current AQI Prediction Model

Models compared: Ridge Regression, Random Forest. Evaluated with RMSE, MAE, and R².

```
Rows usable for current AQI model: 84
Train rows: 67 | Test rows: 17
[Ridge Regression] RMSE=2.51 MAE=2.03 R2=0.992
[Random Forest] RMSE=2.99 MAE=1.92 R2=0.989

Best model: Ridge Regression
Saved to C:\Users\LENOVO\Desktop\AQI_Project\models\model_current.pkl
ProvenanceWarning: Model schema cannot not be inferred without both the feature view and the training dataset version.
Uploading model files: 17%|██████████| 1/6 [00:02<00:13, 2.71s/it]Moving model files from
ry... This is the default behavior. Set keep_original_files=True to copy files instead.
Uploading C:\Users\LENOVO\AppData\Local\Temp\tmpdhkwd5bm\model.pkl: 100.000%|██████████| 1654/1654 elapsed<00:02 remaining<00:00
Model export complete: 100%|██████████| 6/6 [00:13<00:00, 2.33s/it]
Model created, explore it at https://eu-west.cloud.hopsworks.ai:443/p/43137/models/aqi_model_current/7
[hopsworks] registered aqi_model_current
2026-09-05 13:01:13,119 INFO: Closing external client and cleaning up certificates.
2026-09-05 13:01:13,138 INFO: Connection closed.
```

Note on this model's sample size: the current-AQI model can only train on rows that have complete weather data, which (as explained in Section 5.2) only comes from the live hourly pipeline, not the historical backfill. Early in the project this meant very few usable rows. Since the hourly GitHub Actions workflow has been running continuously, this dataset grows by roughly 3 rows per hour (one per city), so this model's sample size - and result quality - improves the longer the automated pipeline has been running before the final training run.

7.2 3-Day Forecast Models

Models compared: Ridge Regression, Random Forest, evaluated against a naive baseline (assume AQI doesn't change) for each of the three horizons.

Horizon	Test rows	Naive RMSE	Ridge RMSE	RF RMSE	Best model	Improvement vs. naive
24h	10,146	69.12	54.34	51.22	Random Forest	25.9%
48h	10,131	72.15	56.71	52.16	Random Forest	27.7%
72h	10,116	71.80	56.99	52.44	Random Forest	27.2%

Random Forest achieved the lowest RMSE for **all three forecast horizons**, making it the best-performing model. Compared with the naive baseline, it reduced RMSE by approximately **25.9% for 24 hours, 27.7% for 48 hours, and 27.2% for 72 hours**. This shows that the trained models are able to capture useful patterns from the available pollution and time-related features for future AQI prediction.

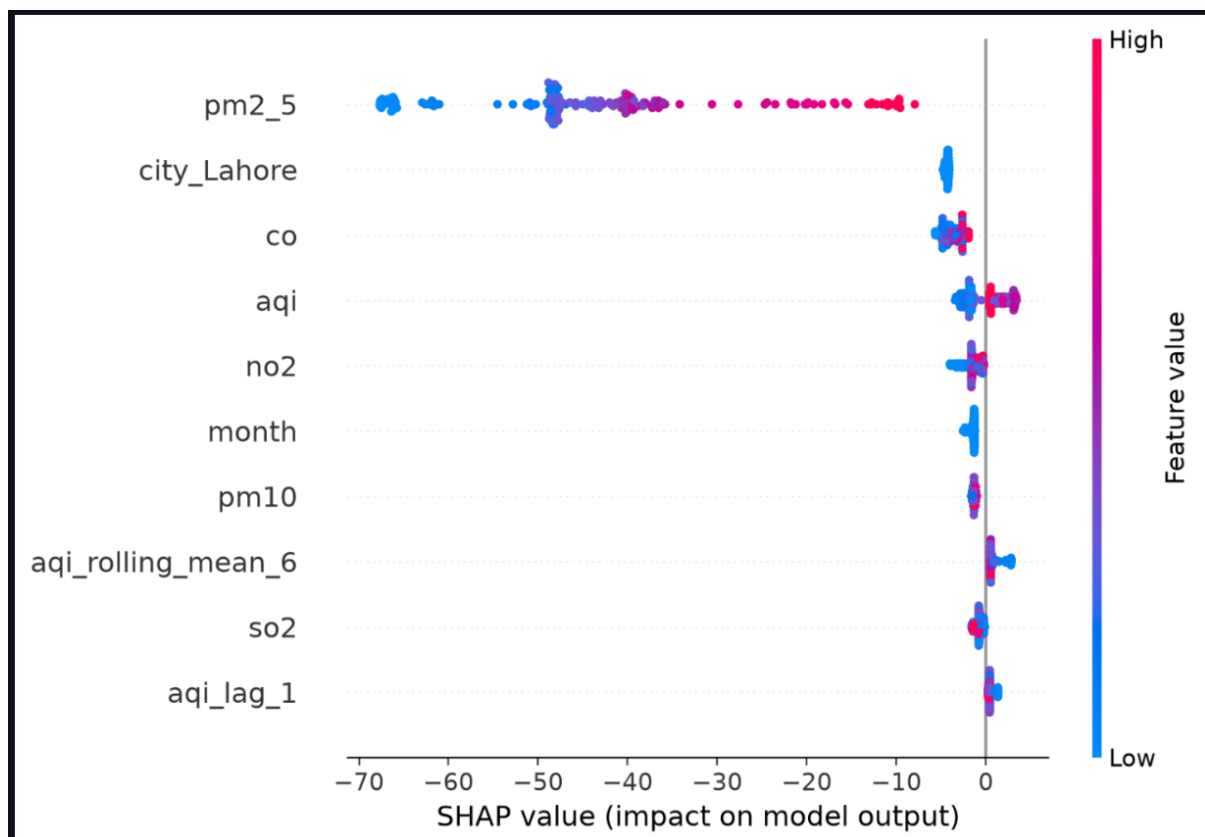
The R² scores for the Random Forest models were **0.145, 0.115, and 0.107** for the 24h, 48h, and 72h horizons respectively. These values indicate that the models capture some predictive signal, although longer-term AQI

forecasting remains challenging. The negative R^2 values of the naive baseline (**-0.556**, **-0.693**, and **-0.686**) also show that simply assuming AQI will remain unchanged performs poorly on the held-out test period.

7.3 Shap Explainability

SHAP (SHapley Additive exPlanations) is used to understand why the AQI model makes a particular prediction. It shows how each input feature contributes to increasing or decreasing the predicted AQI. In the dashboard, the feature-importance chart helps identify which factors, such as PM2.5, CO, PM10, and other weather or pollutant features, have the greatest influence on the model's 24-hour AQI prediction.

This makes the model more transparent and interpretable, allowing users to understand not only the predicted AQI but also the main factors behind that prediction.



8. Automation

Two GitHub Actions workflows keep the system running without needing a laptop to stay on:

- Hourly workflow — runs `feature_pipeline.py`, fetches fresh data for all 3 cities.
- Daily workflow — runs `train_current.py` and `train_forecast.py`, retrains all models on the latest data.

API keys are stored as GitHub Secrets, never committed to the repository.

GitHub Actions tab showing successful runs

38 workflow runs		Workflow ▾	Event ▾	Status ▾	Branch ▾	Actor ▾
✓	Daily Training Pipeline Daily Training Pipeline #8: Scheduled	main		34 minutes ago 4m 13s	...	
✓	Hourly Feature Pipeline Hourly Feature Pipeline #30: Scheduled	main		Today at 08:36 1m 3s	...	
✓	Hourly Feature Pipeline Hourly Feature Pipeline #29: Scheduled	main		Today at 04:03 1m 6s	...	
✓	Hourly Feature Pipeline Hourly Feature Pipeline #28: Scheduled	main		Today at 01:42 1m 14s	...	
✓	Hourly Feature Pipeline Hourly Feature Pipeline #27: Scheduled	main		Sep 3, 22:38 GMT+5 1m 13s	...	
✓	Hourly Feature Pipeline Hourly Feature Pipeline #26: Scheduled	main		Sep 3, 18:22 GMT+5 1m 15s	...	
✓	Hourly Feature Pipeline Hourly Feature Pipeline #25: Scheduled	main		Sep 3, 13:42 GMT+5 1m 15s	...	
✓	Daily Training Pipeline Daily Training Pipeline #7: Scheduled	main		Sep 3, 12:47 GMT+5 4m 13s	...	
✓	Hourly Feature Pipeline	main		Sep 3, 08:35 GMT+5	...	

9. Dashboard

- **Current AQI Monitoring:** Displays the latest AQI values for Sukkur, Karachi, and Lahore.
- **AQI Classification:** Categorizes air quality into levels such as Good, Moderate, Unhealthy, Very Unhealthy, and Hazardous.
- **City-wise Analysis:** Allows users to select a city and view its detailed air-quality information.
- **Pollutant Monitoring:** Displays current PM2.5, PM10, CO, NO, NO₂, O₃, SO₂, and NH₃ measurements.
- **Historical AQI Trends:** Shows how AQI has changed over time using historical data.
- **3-Day AQI Forecast:** Provides predicted AQI values for 24, 48, and 72 hours ahead.
- **Model Performance:** Displays the selected model's RMSE, MAE, and R² metrics.
- **Feature Importance:** Shows which features have the greatest influence on the 24-hour AQI forecast.
- **Pollutant Trends:** Allows users to select individual pollutants and visualize their historical trends.
- **City Comparison:** Compares AQI trends of Sukkur, Karachi, and Lahore on the same chart.
- **Automated Data Updates:** The dashboard uses data continuously collected through the project's automated feature pipeline.

Dashboard Overview



3-Day Forecast



App Streamlit URL: <https://pearlsaqipredictorbk.streamlit.app/>

10. Limitations

- Free-tier historical weather data only goes back about a week, limiting how much weather-complete data the current-AQI model can train on.
- The 3-day forecast doesn't use real future weather forecasts, so it relies on past AQI patterns — accuracy is expected to drop the further out the forecast goes.
- Training data currently spans about 2 months; more history would help the model learn seasonal patterns.
- Only Ridge Regression and Random Forest were compared. More advanced models (XGBoost, LSTM) were listed as optional in the project brief and weren't required for this version.

11. Future Improvements

- Will try XGBoost or a simple LSTM and compare against the current models.

- Will let the hourly pipeline keep growing the dataset over time and retrain on a larger history.
- Will add SHAP-based explanations for individual predictions.

12. Conclusion

This project builds a working AQI prediction system for three Pakistani cities, with a clear separation between predicting current AQI from live readings and forecasting future AQI from historical trends — a distinction that matters for avoiding data leakage and was a deliberate design decision, not an afterthought. The pipeline collects data automatically, stores it in Hopsworks, retrains itself daily, and serves predictions through a dashboard covering all the views specified in the project brief. The results section above is left for the actual run output, and the limitations section is written honestly rather than glossed over, since a project report should reflect what was actually built and tested.

Appendix

A. Repository

GitHub repository URL: https://github.com/BharatKumar292/aqi_predictor

B. Cities Tracked

Sukkur, Karachi, Lahore.